Absolute values on $\mathbb{Q}$

An *absolute value* on $\mathbb{Q}$ is a function $|\cdot|_* : \mathbb{Q} \to \mathbb{R}$ such that:

1. $|0|_* = 0$ and $|x|_* > 0$ for $x \neq 0$.
2. $|\cdot|_* : U(\mathbb{Q}) \to \mathbb{R}^+$ is a group homomorphism. Here $U(\mathbb{Q}) = \mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$ is the group of invertible elements of $\mathbb{Q}$.
3. $|x + y|_* \leq |x|_* + |y|_*$. This is called the triangle inequality.

The first and third conditions ensure that $d(x, y) = |x - y|_*$ defines a metric on $\mathbb{Q}$. We shall return to this point later.

Note that, by the second condition, we have

$$|-1|_*^2 = |1|_* = |1|_*^2$$

By the first condition, it follows that $|-1|_* = |1|_* = 1$.

By the second condition, it follows that $|-x|_* = |x|_*$.

By the second condition, we see that if $x = p/q$ where p and q are integers with $q \neq 0$, then $|x|_* = |p|_*/|q|_*$.

Hence, such an absolute value is *determined* by what values it takes on the set $\mathbb{N}$ of counting numbers.

We note that the third condition says that $|a|_* \leq a$ for each element $a \in \mathbb{N}$. (Prove this by induction on a.)

If $|n|_* = 1$ for all natural numbers, this *is* an absolute value. We will *exclude* this as a *trivial* absolute value. The resulting metric in $\mathbb{Q}$ is the discrete metric and is not particularly interesting.

Non-Archimedean case

Suppose that there is *some* $a \in \mathbb{N}$ such that $a > 1$ and $|a|_* \leq 1$.

Every natural number b can be "expressed in base a" as $b = (c_m \dots c_0)_a$ where $0 \leq c_k < a$ are the "a-digits". This amounts to an expression $b = \sum_{k=0}^{m} c_k a^k$. By the triangle inequality, we see that

$$|b|_* \leq \sum_{k=0}^{m} |c_k|_* |a^k|_*$$

Now, $|a^k|_* = |a|_*^k \leq 1$ and $|c_k|_* \leq c_k < a$. Thus, we see that $|b|_* < (m + 1)a$. (*Warning*: Do not be misled into thinking that $c_k < a$ means that $|c_k|_* \leq |a|_*$. This need not be true as we shall see!)

As is usual in base a representations of numbers, we assume that $c_m \neq 0$. This means that $a^m \leq b < a^{m+1}$. This yields $m \log(a) \leq \log(b) < (m + 1) \log(a)$. (Here the base of the logarithm is irrelevant!). We thus obtain the inequality

$$|b|_* < \left(\frac{\log(b)}{\log(a)} + 1 \right) a$$

Applying this to b^n for large n and using $\log(b^n) = n\log(b)$, we see that

$$|b|_*^n = |b^n|_* < \left(n\frac{\log(b)}{\log(a)} + 1\right) a$$

So, $n \mapsto |b|_*^n$ grows *at most* linearly with n. This forces $|b|_* \leq 1$.

In summary, we have shown the following.

> If $|a|_* \leq 1$ for *some* natural number $a > 1$, then $|b|_* \leq 1$ for *all* natural numbers b.

(Question: How did we use the condition that $a > 1$?)

Since the absolute value is not trivial, we know that there *is* a natural number n such that $|n|_* \neq 1$. By the above condition, this would mean that $|n|_* < 1$.

Writing $n = p_1^{a_1} \cdots p_k^{a_k}$ we see that $|n|_*$ is the product of $|p_i^{a_i}|_*$. It follows that *at least* one prime p has the property that $|p|_* < 1$.

We claim that there is a *unique* such prime p. The proof will be by contradiction.

Suppose that there are two primes $p \neq q$ such that $|p|_* < 1$ and $|q|_* < 1$.

By choosing suitable powers e and f we can assume that $|p|_*^e < (1/2)$ and $|q|_*^f < (1/2)$.

Since p^e and q^f are co-prime, we can write $1 = xp^e - yq^f$ for some natural numbers x and y. We now obtain

$$1 = |1|_* \leq |x|_*|p^e|_* + |-y|_*|q^f|_* < (|x|_* + |y|)) \frac{1}{2} \leq 1$$

since x and y are natural numbers, so $|x|_*, |y|_* \leq 1$. This is a contradiction.

(Question: Try to form a proof that does *not* use proof by contradition. Suppose p is a prime such that $|p|_* < 1$ and q is any other prime. Use calculations like those above to show that $|q|_* \geq 1$.)

Thus, we have shown the following:

> There is a *unique* prime p such that $|p|_* < 1$. For all other primes q, we have $|q|_* = 1$.

Given *any* rational number $r \neq 0$, we write $r = p^k(m/n)$, where k and m are integers, $n > 0$ is a natural number *and* p does not divide m or n. From the above, it follows that $|r|_* = |p|_*^k$.

Note that so far, we have *not* shown there *is* such an absolute value!

It is clear that the definition $|r|_* = |p|_*^k$ satisfies the second condition. By taking $|0|_* = 0$, and $|p|_*$ to be a positive real number, we see that the first condition is also satisfied. We now claim that for *any* $0 < \alpha < 1$, if we put $|p|_* = \alpha$, the third condition is also satisfied.

The triangle inequality is clear when x or y is 0. Thus, we may assime that $x = r = p^k(m/n)$ as above with $k \geq 0$ and we only need to prove $|r+1|_* \leq |r|_*+1$.

Since $n \neq 0$, we need only prove $|p^k m + n|_* \leq |p^k m|_* + |n|_*$. Since $\gcd(m, p) = \gcd(n, p) = 1$. This is reduced to proving $|p^k m + n|_* \leq |p^k|_* + 1$ for $k \geq 0$ and m, n integers.

In fact, $p^k m + n$ is an integer, so we know that $|p^k m + n|_* \leq 1$. Thus, the inequality follows.

Note that the set S_k consisting of rational numbers r of the form $p^k(m/n)$ where k, m and n are integers with $n \neq 0$ and $\gcd(m, p) = 1 = \gcd(n, p)$ is precisely the set of numbers of absolute value α^k in this norm. It follows that the open unit ball of radius ϵ around 0 in $\mathbb{Q}$ is precisely the union of S_k for k such that $\alpha^k < \epsilon$. This is all k larger than $-\log_{1/\alpha}(\epsilon)$. Thus, we see that the *topology* on $\mathbb{Q}$ is unchanged if we replace α by a β such that $0 < \beta < 1$.

Thus, we make the convenient choice $\alpha = 1/p$ which keeps us informed of which prime we are working with!

> The p-adic norm $|\cdot|_p$ on $\mathbb{Q}$ is defined, for a prime p, by defining $|p|_p = 1/p$ and $|q|_p = 1$ for q a prime different from p.

One can use the above calculation to show that:

$$|x + y|_p \leq \max\{|x|_p, |y|_p\}$$

for all x and y in $\mathbb{Q}$. In fact, if $|x|_p < |y|_p$, then $|x + y|_p = |y|_p$.

Archimedean case

The opposite case to the one above is when $|a|_* > 1$ for all $a \in \mathbb{N}$ with $a > 1$.

Given a natural number b, as above we use $b = \sum_{k=0} c_k a^k$ with $0 \leq c_k < a$ to get

$$|b|_* \leq \sum_{k=0}^{m} |c_k|_* |a^k|_* \leq (m+1)a|a|_*^m$$

As above, we note that $m \leq \log_a(b)$. We now apply this to b^t to get

$$|b|_*^n \leq (n\log_a(b) + 1)\, a|a|_*^{n\log_a(b)}$$

Taking n-roots (of positive real numbers), we get

$$|b|_* \leq (n\log_a(b) + 1)^{1/n}\, a^{1/n} |a|_*^{\log_a(b)}$$

Now, one checks that for *any* positive constants C and D we have

$$\lim_{n\to\infty} (nC + D)^{1/n} = 1$$

It folows that $|b|_* \leq |a|_*^{\log_a(b)}$. Equivalently,

$$|b|_*^{1/\log(b)} \leq |a|_*^{1/\log(a)}$$

Since this is true for *all* integers $a > 1$ and $b > 1$, we see that

$$|b|_*^{1/\log(b)} = |a|_*^{1/\log(a)}$$

In other words, we have a suitable positive constant α, such that for all $b > 1$ integer, we have $|b|_* = \alpha^{\log(b)}$. Writing $\alpha = e^\beta$ we see that

$$|b|_* = b^\beta \text{ for all integers } b > 1$$

It follows easily that $|b|_* = |b|_\infty^\beta$ for *all* integers b; here $|b|_\infty$ denotes the usual absolute value of a rational number.

By the earlier discussion we see that $|r|_* = |r|_\infty^\beta$ for *all* rational numbers r.

Note that, in order that the triangle inequality to be satisfied we must have $\beta > 0$.

The open unit ball of radius ϵ around 0 in $\mathbb{Q}$ with respect to $|\cdot|_*$ is the same as the open unit ball of radius $\epsilon^{1/\beta}$ around 0 with respect to $|\cdot|_\infty$. It follows easily that the topology associated with these two metrics on $\mathbb{Q}$ is the same. Thus, we again make the convenient choice, which is $\beta = 1$.

Ostrowski's Theorem

A non-trivial absolute value $|\cdot|_*$ on $\mathbb{Q}$ can be of two types:

- It satisfies $|n|_* \leq 1$ for all integers n.
- It satisfies $|n|_* > 1$ for all integers n.

In the first case, there is a prime p so that $|\cdot|_*$ is equivalent to $|\cdot|_p$.

In the first case, $|\cdot|_*$ is equivalent to $|\cdot|_\infty$.

Further details can be found on the relevant page of Wikipedia.

Completion

Given an absolute value $|\cdot|_*$ on $\mathbb{Q}$, we have a metric space structure on $\mathbb{Q}$ such that the distance is given by $d(x, y) = |x - y|_*$. The first and third conditions on the absolute value ensure that this is a metric.

We can thus form the *complete* metric space $\mathbb{Q}_*$ elements of which are equivalence classes of Cauchy sequences (x_n) with respect to the metric d. The equivalence relation is defined by

> $(x_n) \sim (y_n)$ if, for all integers $k > 1$, there is an $N(k)$ such that $d(x_n, y_n) < (1/k)$ if $n > N$

It is worth noting at this point that if $d(x, y)$ is replaced by a *positive* power of it, then we only need to replace k by another (possibly larger) integer K. Thus, equivalent absolute values in the sense introduced above give rise to equivalent notions of Cauchy sequences.

Another useful observation is that in a metric space, if a is any *fixed* point $d(x, a)$ is a uniformly continuous function of x. Hence, the image $(d(a, x_n))$ of a Cauchy sequence (x_n) is a Cauchy sequence of real numbers. Hence, $(d(a, x_n))$ is *uniformly* bounded. In particular, if (x_n) is a Cauchy sequence, then $|x_n|_* = d(x_n, 0)$ is uniformly bounded.

Furthermore, if (x_n) is a Cauchy sequence *inequivalent* to the constant sequence (0), then $(d(0, x_n))$ is a Cauchy sequence of reals that *does not* converge to 0. It follows that there are integers $k > 1$ and an N such that $|x_n|_* > 1/k$ for all $n >\leq N$.

We now wish to see that $\mathbb{Q}_*$ is also a field; it has elements corresponding to 0 and 1 and the operations of subtraction and division by non-zero elements make sense and have the usual properties.

Given Cauchy sequences (x_n) and (y_n), we can define the sequence $(z_n) = (x_n - y_n)$. We have

$$
\begin{aligned}
d(z_m, z_n) = d(x_m - y_m, x_n - y_n) = |(x_m - y_m) - (x_n - y_n)|_* \\
\leq |x_m - x_n|_* + |y_m - y_n|_* = d(x_m, x_n) + d(y_m, y_n)
\end{aligned}
$$

This can be used to show that (z_n) is a Cauchy sequence. Given that (w_n) is another Cauchy sequence and we define $(t_n) = x_n - y_n + w_n$ and check that

$$
d(t_m, t_n) \leq d(x_m, x_n) + d(y_m, y_n) + d(w_m, w_n)
$$

This can be used to show that (t_n) is a Cauchy sequence. If $(y_n) \sim (w_n)$ are equivalent Cauchy sequences, this can be used to show that (t_n) and (x_n) are equivalent Cauchy sequences by using the fact that $(x_n) = (t_n - w_n + y_n)$ and the above inequality with the roles of x and t interchanged.

This shows that the difference of equivalence classes of Cauchy sequences is well-defined. This leads to a definition of addition. The associativity property of addition is also easily checked.

Given that (y_n) is a Cauchy sequence which is *not* equivalent to 0, we saw that there is a $k > 1$ and an N such that $|y_n|_* > 1/k$ for all $n \geq N$. We then replace (y_n) by the equivalent sequence (z_n) which satisfies $z_n = y_N$ for $n < N$ and $z_n = y_n$ for $n \geq N$. Since all elements of the sequence (z_n) are non-zero, we can now divide by these elements.

We can then define the sequence $(w_n) = (x_n/z_n)$. We have

$$\begin{aligned} d(w_m, w_n) = |(x_m/z_m) - (x_n/z_n)|_* \\ = \left| \frac{z_n(x_m - x_n) + x_n(z_m - z_n)}{z_n z_m} \right|_* \\ \leq \frac{|z_n|_* d(x_m, x_n) + |x_n|_* d(z_m, z_n)}{|z_m z_n|_*} \end{aligned}$$

Since (z_n) is a Cauchy sequence $|z_n|_*$ is *uniformly* bounded by some positive number A. Similarly, $|x_n|_*$ is uniformly bounded by some positive number B. Finally $|z_n z_m|_* > 1/k^2$, so we get

$$d(w_m, w_n) \leq k^2 \left(Ad(x_m, x_n) + Bd(z_m, z_n) \right)$$

which can be used to show that (w_m) is a Cauchy sequence; let us call it the ratio of two Cauchy seqences and denote it as $(x_n)/(y_n)$. Similar arguments as above can now be used to show that the ratio $(x_n)/(y_n)$ respects the equivalence relation on Cauchy sequences and has all the usual properties.

Further details can be found on the relevant page of ProofWiki.